CHRISTOPHER MIKAL LOGAN

AI/ML Engineer & Architect | Generative AI, RAG & LLM Systems | Applied Machine Learning Leadership

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PROFESSIONAL SUMMARY

AI/ML engineer and architect with 30+ years of software engineering depth, now focused on building and shipping production Generative AI systems. Hands-on across the full modern LLM stack: Retrieval-Augmented Generation (RAG), multi-model orchestration, local and cloud inference, vector databases, and embedding pipelines. Led enterprise Generative AI engineering at BNY Mellon, delivering a custom RAG-powered agent and an enterprise GitHub Copilot rollout under an OpenAI partnership. Builds independently on Apple Silicon with Ollama, LangChain, and ChromaDB, and ships AI-enhanced applications end to end. Combines applied ML practice with the architecture, leadership, and compliance experience (NIST, FDA, SOX, NACHA) to take AI systems from prototype to dependable production deployment.

AI / ML TECHNICAL SKILLS

LLMs & Generative AI: RAG architecture, multi-LLM orchestration, prompt engineering, agent design, self-service augmentation layers, LLM evaluation & test harnesses, OpenAI API, GitHub Copilot, Llama 3, Mistral

Frameworks & Tooling: LangChain, Ollama, n8n workflow automation, Hugging Face ecosystem, PyTorch, TensorFlow

Vector & Embeddings: ChromaDB (vector store), nomic-embed-text embeddings, semantic search, document chunking & retrieval pipelines

ML Engineering: Neural network design, OCR modernization, model integration into firmware & applications, real-time inference, MLOps & deployment patterns

Languages: Python (primary for AI/ML), C++, Java, JavaScript/TypeScript, SQL (PL/SQL, T-SQL), Lisp, Bash

Cloud & Infrastructure: AWS (EC2, S3, CloudFront), Akamai CDN/NetStorage, Docker, CI/CD (Jenkins, GitHub Actions), Apple Silicon local inference

Data & Backend: PostgreSQL, Oracle/PL-SQL, DB2, MySQL, REST APIs, Spring Boot, Node.js, high-availability systems

Governance & Compliance: NIST frameworks, FDA / SaMD, SOX, NACHA, model governance, IP & USPTO filing

SELECTED AI / ML PROJECTS

Lisp Project - AI-Native Civilizational Evaluation Platform

Personal / Open Source | github.com/pedalshoe

- Built a Common Lisp based hybrid reasoning system that ingests live global development data and evaluates nation-scale alignment against an ethical governance framework. Integrated the Lisp core with TypeScript and React/Next.js components to support AI/LLM style workflows such as retrieval grounded evaluation, symbolic scoring, and interpretable decision support.
- Implemented a modular Lisp evaluation engine with normalized ingestion interfaces, multi-source adapter contracts, profile packaging, and rule-based scoring across 38 canonical metric slots. Designed the system to merge structured evidence from sources such as WHO, UNICEF, World Bank, UNESCO UIS, ILOSTAT, FAOSTAT, and UN Women into a transparent scoring pipeline.

Local RAG Agent Stack on Apple Silicon

Personal / Self-directed

- Built a fully local Retrieval-Augmented Generation stack running on Apple Silicon using Ollama for inference, LangChain for orchestration, and ChromaDB as the vector store, with nomic-embed-text for embeddings.
- Designed the document ingestion, chunking, embedding, and semantic-retrieval pipeline end to end, enabling private, offline LLM querying over a custom knowledge base with no external API dependency.

AI-Enhanced Kitchen Display System (KDS) - Multi-LLM Pipeline

Worlwin LLC | github.com/pedalshoe (kds-demo)

- Engineered a multi-model Ollama pipeline routing tasks across specialized LLMs - Llama 3 for order extraction and Mistral for RAG-powered Q&A - orchestrated with n8n workflow automation.
- Implemented order-extraction logic and an LLM test-case suite, achieving a 60% performance gain in chatbot interaction, and produced production deployment guidance for the stack.

PROFESSIONAL EXPERIENCE

Sr. Vice President / Sr. Principal Developer - Generative AI Engineering

Bank of New York Mellon (BNY Mellon) | New York, NY | Feb 2001 – Jan 2025

- Early adopter of the firm's Generative AI engineering efforts, designing and deploying a custom LLM-powered agent as an augmentation layer for business-specific workflows - enabling a self-service model that eliminated centralized bottlenecks.
- Applied Retrieval-Augmented Generation (RAG) to ground responses in domain-specific data, improving accuracy and relevance for internal knowledge queries.
- Facilitated enterprise-wide GitHub Copilot integration across development teams and managed a strategic partnership with OpenAI to expand the firm's AI engineering capabilities and adoption.
- Led an OCR modernization initiative using neural networks (PyTorch/TensorFlow), significantly improving document recognition accuracy over legacy rule-based systems.
- Designed a firm-wide reporting system that reduced batch processing time by 93% (from 8–16 hours to 1 second–2 hours), and architected a Content Delivery Platform that migrated 100+ applications to Akamai CDN, offloading 95–99% of server traffic.
- Managed engineering teams of 5 to 50+ across AI, platform, and mobile programs; owned architecture, delivery, and 24/7 production operations.

Founder / Lead Developer / Fractional CTO

Worlwin LLC | New York, NY (Remote) | Jan 2009 – Present

- Build AI-enhanced applications for clients, including the multi-LLM KDS pipeline (RAG, n8n, multi-model Ollama) and neural-network OCR modernization with PyTorch/TensorFlow.
- Provide fractional C-level technology leadership - AI/ML strategy, architecture, technology selection, and team building - across finance, healthcare, government, and media clients.
- Deliver full-stack platforms end to end: payment integration, mobile apps, Angular/D3.js data visualization, REST APIs, and high-availability database clustering.

Senior Software Engineer - Embedded AI / Medical Devices

Burleon Tech | Remote | Jan 2021 – Present

- Architect and integrate AI/ML capabilities directly into ARM-based embedded medical device firmware, using Python with real-time sensor data acquisition over GPIO.
- Navigate FDA / Software-as-a-Medical-Device (SaMD) compliance and manage patent / IP processes for proprietary algorithms and device innovations.

Software Engineer - Optics & Adaptive Systems

Raytheon Systems Company | El Segundo, CA | Jan 1997 – Jan 2001

- Developed adaptive-optics control software in C++ on UNIX (one of two senior engineers maintaining 500K+ lines), plus Java-to-C socket IPC bridging a VxWorks RTOS, and Hubble Space Telescope data analysis.

Member of Technical Staff - Defense Systems

Technology Service Corporation (TSC) | Silver Spring, MD | Jun 1994 – Dec 1996

- Built radar/sensor analysis software and data-visualization tooling in C++ on UNIX; performed sensor modeling with MATLAB and Xpatch and constructed model databases from USGS datasets.

EDUCATION

M.S., Computer Science - Pace University, New York, NY
B.S., Computer Science - Andrews University, Berrien Springs, MI

PROFESSIONAL DEVELOPMENT

- GitHub Copilot - Enterprise Deployment Lead (BNY Mellon, 2023)
- OpenAI Strategic Partnership - AI Competency Development (BNY Mellon, 2023)
- USPTO Trademark & IP Registration - End-to-End Practitioner